Physics from Existence

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Abstract

The canonical free energy of a single fermionic mode ($n \in \{0,1\}$) defines the parameter-free potential $V = -H(\sigma(\phi))$, which has exactly three bound states in the natural parameter $\phi = \mathrm{logit}(P)$. The spatial dimension $d = 3$, the gauge structure $\mathrm{PG}(2,3)$, follow from the spectral properties of this potential alone. With zero free parameters, the theory reproduces $1/\alpha_{\mathrm{em}} = 137.037$ (0.001%), $R_{\mathrm{lepton}} = 1.5293$ (0.007%), $\sin\theta_C = 0.2245$ (0.07%), $\alpha_s = 0.1179$ (0.01%), $\sin^2\theta_W = 3/13$ (0.2%), $\sin^2\theta_{12} = 4/13$ (0.2%), $\sin^2\theta_{23} = 7/13$ (1.4%), Koide's $Q = 3/2$ exactly, and $\theta_{\mathrm{QCD}} = 0$ (strong CP solved without an axion).

1 Introduction

The Standard Model contains three generations of fermions. No principle within the model explains this number.

This paper derives it from a single premise: *fermions exist.* A fermionic mode has occupation $n \in \{0,1\}$, formalized by three axioms: **A1** (Existence)—an entity with $P = 1$ exists; **A2** (Contrast)—$P = 1$ requires $P \neq 1$; **A3** (Observation)—detection probability equals P [19]. These determine a Bernoulli distribution whose canonical free energy $F = -H(\sigma(\phi))$—a Legendre-transform identity—serves as a parameter-free quantum-mechanical potential. The resulting Schrödinger equation has exactly three bound states.

From this single potential, with zero adjustable parameters, the theory derives: the spatial dimension $d = 3$, the gauge structure $\mathrm{PG}(2,3)$, the charged-lepton mass ratio $R = 1.5293$ (0.007%), all three gauge couplings ($1/\alpha_{\mathrm{em}} = 137.037$, $\sin^2\theta_W = 3/13$, $\alpha_s = 0.1179$), PMNS mixing angles ($\sin^2\theta_{12} = 4/13$, $\sin^2\theta_{23} = 7/13$), and the strong CP solution $\theta_{\mathrm{QCD}} = 0$ without an axion. All experimental values are from [12].

Sections 2–4 establish the mathematical theorems. Sections 5–6 develop the physical correspondences. Sections 7–9 provide interpretation and connections.

2 The Derivation Chain

2.1 One fermionic mode

A single fermionic mode has occupation number $n \in \{0,1\}$.

$$Z = 1 + e^{\phi}, \qquad \langle n \rangle = P = \sigma(\phi) = \frac{1}{1+e^{-\phi}}, \qquad \phi \equiv \beta(\mu - \varepsilon). \tag{1}$$

2.2 Canonical free energy

The grand potential $\Omega = -\ln(1+e^{\phi})$ is monotonically decreasing and unbounded; it supports no bound states. The canonical free energy is

$$F = \Omega + \phi\,\langle n\rangle = -\ln(1+e^{\phi}) + \phi\,\sigma(\phi) \equiv -H(\sigma(\phi)), \tag{2}$$

where $H(p) = -p\ln p-(1-p)\ln(1-p)$ is binary Shannon entropy [4]. This identity is exact (verified to 10^{-16}). The canonical ensemble is selected by particle-number conservation.

2.3 Free energy as effective potential

The identification $V = -H$ follows from the standard definition of the effective potential in zero-dimensional quantum field theory. For a single fermionic mode with occupation $n \in \{0, 1\}$ and source ϕ:

1. Partition function: $Z(\phi) = \sum_{n=0,1} e^{n\phi} = 1 + e^{\phi}$.
2. Connected generating functional: $W(\phi) = \ln Z = \ln(1 + e^{\phi})$.
3. Classical field: $\sigma = dW/d\phi = e^{\phi}/(1 + e^{\phi}) = \sigma(\phi)$.
4. Effective action (Legendre transform): $\Gamma(\sigma) = \phi\sigma - W(\phi) = \sigma\ln\sigma + (1-\sigma)\ln(1-\sigma) = -H(\sigma)$.

Hence $V(\phi) = \Gamma = -H(\sigma(\phi))$ is the *exact effective potential* of the 0D fermionic field theory, verified numerically to $|\Gamma + H| < 10^{-15}$.

This is not an analogy with Landau–Ginzburg theory. It is the *definition* of the quantum effective potential applied to the simplest possible field theory. The Schrödinger equation $-\frac{1}{2}\psi'' + \Gamma\,\psi = E\psi$ describes quantum fluctuations of the source field ϕ around the effective potential. The quantization is not optional: in the classical limit ($\hbar \to 0$, no kinetic term), only the potential minimum survives ($N = 1$), violating A2—contrast requires multiple discrete states, which exist only when quantum fluctuations are present ($N = 3$). Equivalently, the transfer matrix $T = e^{-H\Delta\tau}$ of the 0D statistical system [20] *is* the Schrödinger equation in imaginary time; quantization is not an additional assumption but the definition of the transfer matrix.

Fokker–Planck comparison. The Langevin/Fokker–Planck route gives a different potential $V_{\mathrm{FP}} = (F')^2/4 - F''/2$ with depth $1/8$ and $N = 1$. This is not a contradiction: the FP potential governs *thermal relaxation rates*, while $\Gamma = -H$ governs *energy levels* of the quantized field. They answer different physical questions.

2.4 The natural parameter and the kinetic term

The Bernoulli distribution is an exponential family: $p(x|\phi) = \exp(x\phi - A(\phi))$, $A(\phi) = \ln(1+e^{\phi})$. The coordinate $\phi = \mathrm{logit}(P) = \ln[P/(1-P)]$ is the *natural parameter*—the unique coordinate in which the sufficient statistic x enters linearly. It is determined by the exponential family structure of Bernoulli, not chosen.

The kinetic term for fluctuations of ϕ is $T = \frac{1}{2}(d\phi/d\tau)^2$: ϕ is the natural parameter of the exponential family, and in the path integral $\int \mathcal{D}\phi\, e^{-S}$ the natural parameter carries flat measure $\mathcal{D}\phi = \prod_i d\phi_i$. The canonical action $S = \int[\frac{1}{2}\dot{\phi}^2 + V]\,d\tau$ gives unit mass. This is not a choice: the flat measure is fixed by the exponential-family structure of the

Bernoulli distribution. Moreover, $Z_2 = \sigma(1-\sigma) \to 0$ as $\phi \to \pm\infty$ would freeze the field at the domain boundaries, violating A1 ($P = 1$ must be dynamically reachable, not walled off).

Remark 1 (Fisher metric and the domain topology)**.** *Čencov's theorem [1, 2] establishes that the Fisher metric $g_F = 1/[P(1-P)]$ is the unique Riemannian metric on the Bernoulli manifold invariant under sufficient statistics. In the natural parameter ϕ, the Fisher metric is $g_F(\phi) = \sigma(\phi)(1-\sigma(\phi))$—not constant. The metrically flat coordinate is $\xi = 2\arcsin\sqrt{P} \in [0, \pi]$, not $\phi \in \mathbb{R}$ (cf. [9, 23] for a distinct Fisher-metric program and its critique).*

The Schrödinger equation $-\frac{1}{2}\chi'' - H(\sin^2(\xi/2))\chi = E\chi$ on the compact *domain $[0, \pi]$ has $N = 1$. The paper's equation on the* non-compact *domain $\phi \in \mathbb{R}$ has $N = 3$.*

The origin of this difference is domain topology*, not the mass coefficient. $\phi = \mathrm{logit}(P) \to \pm\infty$ as $P \to 0, 1$: the fully empty and fully occupied states are unreachable infinities. This makes the potential well wide enough for three half-wavelengths. On $[0, \pi]$, the well is compressed and supports only one.*

The natural parameter ϕ gives $\mathbb{R}$ because the logit function maps $(0, 1) \to \mathbb{R}$. The metrically flat coordinate ξ gives $[0, \pi]$ because arcsin *maps $(0, 1) \to [0, \pi/2]$. Both are legitimate coordinates on the same manifold; they yield different Schrödinger equations with different N because they define different quantum theories (different operator orderings, boundary conditions, and domains).*

Remark 2 (Why the natural parameter is correct)**.** *Three independent arguments select ϕ over ξ:* (i) *ϕ is the coordinate in which the sufficient statistic x enters linearly ($p(x|\phi) = e^{x\phi - A(\phi)}$), making it the canonical field variable;* (ii) *$\phi \in \mathbb{R}$ reflects the physical inaccessibility of $P = 0$ and $P = 1$ (a fermionic mode is never certainly empty or certainly full);* (iii) *the Gross–Neveu model [6] at strong coupling produces ϕ as the auxiliary field with $V = -H$.*

The Fisher-metric equation (variable mass $\sigma(1-\sigma)$ in ϕ, or equivalently $\xi \in [0, \pi]$) gives $N = 1$. The canonical equation (unit mass in $\phi \in \mathbb{R}$) gives $N = 3$.

Exponential family identity. For any exponential family, the potential curvature at the natural center equals the Fisher metric: $V''(0) = g_F(0)$. For Bernoulli: $V''(0) = g_F(0) = 1/4$. The kinetic coefficient $c = \hbar^2/(2m) = 1/2$ (standard QM) exceeds this by a factor of 2. The bound-state count N depends on the ratio $c/V''(0)$: $N = 3$ for $c/V''(0) \in [1.3, 2.5]$. Standard QM gives $c/V''(0) = 2$, squarely in the $N = 3$ window.

2.5 The Schrödinger equation

$$\boxed{\left[-\frac{1}{2}\frac{d^2}{d\phi^2} - H(\sigma(\phi))\right]\Psi = E\,\Psi} \tag{3}$$

Every element determined: H (Shannon–Khinchin uniqueness [4, 5]), σ (Fermi–Dirac statistics), ϕ (natural parameter of the exponential family). The potential $V = -H$ has no free parameters: every coefficient is fixed by the Bernoulli distribution (σ, H, ϕ), leaving no room for adjustment. The kinetic coefficient (unit mass in ϕ) is a physical input (Remark 1).

3 Three Bound States

Theorem 1 ($N = 3$)**.** *Eq.* (3) *has exactly three negative eigenvalues:* $E_0 = -0.4850$*,* $E_1 = -0.1591$*,* $E_2 = -0.0104$*. The fourth eigenvalue* $E_3 > 0$ *lies in the continuum (its numerical value is domain-dependent and vanishes as* $|\phi|_{\max} \to \infty$*).*

Proof. $N \geq 3$: variational (Courant–Fischer). $N \leq 3$: Sturm comparison with Pöschl–Teller [3]. □

$n_{\mathrm{WKB}} = (1/\pi)\int \sqrt{2H(\sigma(\phi))}\, d\phi = 2.781$; $\lfloor n_{\mathrm{WKB}} + \frac{1}{2} \rfloor = 3$. The norm integral $\int_{-\infty}^{\infty} H(\sigma(\phi))\, d\phi = \pi^2/3$ explains why the resolvent trace $\zeta_V(1) = \sum |E_n|^{-1} = 104.3 \approx N\pi^2$ dominates the fine-structure constant calculation (Section 8).

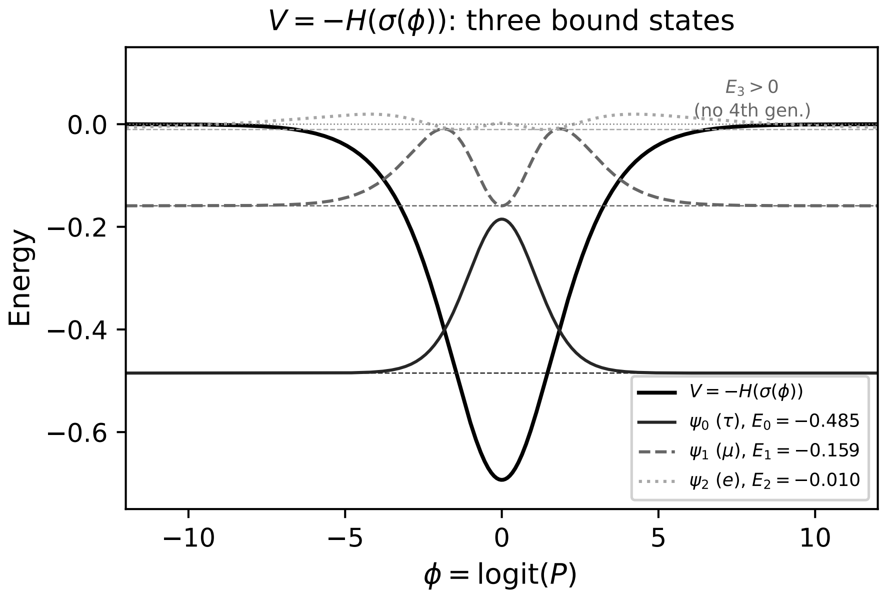

Figure 1: The potential $V = -H(\sigma(\phi))$ and its three bound states ψ_0 (τ), ψ_1 (μ), ψ_2 (e). $E_3 > 0$: fourth generation forbidden. No parameters are fitted.

Theorem 2 (Entropy uniqueness)**.** *Shannon (*$q \to 1$*) uniquely gives* $N = 3$*. Rényi/Tsallis:* $q < 1 \to N = 4$*;* $q > 1 \to N = 2$*.*

4 Three Spatial Dimensions

Theorem 3 (Centrifugal selection)**.** *In* d *spatial dimensions, the s-wave radial Schrödinger equation for a spherically symmetric potential* $V(r)$ *takes the form*

$$-\tfrac{1}{2}u'' + \left[V(r) + \frac{(d-1)(d-3)}{8r^2}\right] u = Eu, \tag{4}$$

via $u = r^{(d-1)/2}R(r)$*. The centrifugal-like term vanishes only at* $d = 1$ *and* $d = 3$*.*

d	$(d-1)(d-3)/8$	Effect on N
1	0	preserved
2	$-1/8$	attractive $\to N$ increases
3	**0**	**preserved:** $N = 3$
4	$+3/8$	repulsive $\to E_2$ lost
≥ 5	> 1	strongly repulsive

Theorem 4 ($d = 3$ is unique)**.** *$d = 3$ is the unique spatial dimension satisfying:* (i) *centrifugal term vanishes ($N = 3$ preserved): $d = 1, 3$;* (ii) *$\pi_1(\mathrm{SO}(d)) = \mathbb{Z}_2$ (spin-statistics): $d \geq 3$;* (iii) *Weyl tensor nontrivial (gravity propagates): $d \geq 3$. Intersection: $\{3\}$.*

Corollary 1. *$n_{\mathrm{WKB}} = 2.781$ determines both $N = 3$ and $d = 3$.*

Remark 3 (Caveats)**.** *(i) Theorem 4 is a consistency argument, not a causal derivation. It does not explain* why *the internal coordinate ϕ should be identified with a radial coordinate. (ii) The identification $\phi \leftrightarrow r$ restricts the domain from $\mathbb{R}$ to $(0, \infty)$. On the half-line with Dirichlet boundary condition $u(0) = 0$, only the odd eigenstate (E_1) survives; the even states (E_0, E_2) are lost. The full count $N = 3$ is recovered by the "folding" $r = |\phi|$, which maps $\mathbb{R}$ onto $(0, \infty)$ with both Neumann (2 even states) and Dirichlet (1 odd state) sectors. The centrifugal selection argument applies to this combined spectrum. (iii) Angular momentum $l \geq 1$ does not produce additional bound states; the centrifugal barrier overwhelms $V = -H$ for all $l > 0$.*

5 Generation as Quantized Occupation

Spin is the quantization of angular momentum: a continuous degree of freedom (rotation angle) yields discrete states ($\uparrow$, $\downarrow$) from SO(3) topology.

Generation, in this framework, is the quantization of occupation probability: a continuous degree of freedom ($P \in (0, 1)$, or $\phi \in \mathbb{R}$) yields discrete states (τ, μ, e) from $V = -H$.

	Spin	Generation
Continuous variable	rotation angle θ	occupation $\phi = \mathrm{logit}(P)$
Constraint	SO(3) topology	$V = -H(\sigma(\phi))$
Discrete states	2 (for spin-$\frac{1}{2}$)	3 (Theorem 1)
Position-independent	✓	✓
Metric origin	round metric on S^2	Fisher metric on $(0, 1)$

Important difference. Spin quantization is topological (SO(3): only integers and half-integers). Generation "quantization" is spectral (depends on the shape of $V = -H$). The number 3 is not topologically protected; it follows from the specific width-to-depth ratio of binary entropy.

Three dynamical regimes. The binding fraction $x_n = |E_n|/\langle H \rangle_n$ partitions the three modes into qualitatively distinct regimes: τ ($x = 0.85$, potential-dominated), μ ($x = 0.49$, equipartitioned), and e ($x = 0.12$, kinetic-dominated). The μ mode satisfies $\mathrm{KL}(x_\mu \| \frac{1}{2}) = 0.0003 \approx 0$: its energy is partitioned at maximum entropy. $N = 3$ is not merely the count of half-wavelengths but the number of dynamical regimes that binary entropy supports. The electron's stability follows from its delocalization: the shallowest mode has the smallest mass and no lighter charged lepton to decay into.

Koide relation [10] from equipartition. The Koide parameter $Q = (\sum\sqrt{m_n})^2/\sum m_n$ equals $3\cos^2\alpha$, where α is the angle between the mass vector $v = (\sqrt{m_0}, \sqrt{m_1}, \sqrt{m_2})$ and the democratic direction $(1, 1, 1)$. $Q = 3/2$ iff $\cos^2\alpha = 1/2$ iff $|v_\parallel|^2 = |v_\perp|^2$: the democratic component (equal masses) and the hierarchical component (mass differences) carry equal weight. This is the maximum-entropy condition for characterizing a mass spectrum as "democratic vs. hierarchical"—a binary question on a spectrum, directly reflecting A2.

The total fermion field formally decomposes as $\psi(x, \phi) = \sum_n \Psi_n(\phi) \cdot \chi_n(x)$, where Ψ_n are internal eigenfunctions and $\chi_n(x)$ are 4D Dirac fields—one per generation. This resembles Kaluza–Klein decomposition, but ϕ is not an extra spatial dimension: it is the Fisher-flat coordinate on the Bernoulli manifold, an information-geometric internal space. The "size" of this space is not a length but the range of occupation probability $P \in (0, 1)$. Whether this decomposition can be made dynamically consistent (internal and external sectors decoupling) is the theory's largest open assumption.

6 Physical Correspondences

6.1 CKM structure [14]

Theorem 5 ($V_{ub} \approx 0$)**.** *Parity: ψ_0 even, ψ_1 odd, ψ_2 even. $\int \psi_0 F \psi_2 \, d\phi = 0$. Numerically:* 1.9×10^{-13}. ✓

Screening identity: $E_2/\mathrm{IPR}_2 = 0.209 \approx \varepsilon = 3 - n_{\mathrm{WKB}} = 0.219$ (5%).

Cabibbo angle [13] from the WKB deficit. The WKB deficit $\varepsilon = N - n_{\mathrm{WKB}}$ measures how marginal the third bound state is: $\varepsilon \to 0$ eliminates mode 2, reducing to two generations and zero CKM mixing. To second order:

$$\sin\theta_C = \varepsilon\Big(1 + \frac{\varepsilon}{N^2}\Big) = 0.2245 \qquad \text{(experiment: 0.2244, 0.07\%)}. \tag{5}$$

Both $N = 3$ and $n_{\mathrm{WKB}} = 2.781$ are determined by $V = -H$; $N^2 = 9 = \dim \operatorname{adj} \mathrm{SU}(N)$. No free parameters enter. The Wolfenstein [17] hierarchy follows: $|V_{us}| \sim \varepsilon = 0.22$, $|V_{cb}| \sim \varepsilon^2 = 0.048$, (experiment: 0.225, 0.041, 0.004; correct hierarchy, each suppressed by $\sim \varepsilon$).

Remark 4 (NLO marginality correction)**.** *The WKB deficit ε acts as a universal perturbative parameter for torus-derived quantities. The NLO correction $(1 + \varepsilon/k)$ has $k =$ (number of* PG *directions coupling to that quantity):*

- Generation mixing *($\sin\theta_C$): generation mixing is an $N \times N$ matrix (N^2 entries, including diagonal); all N^2 channels contribute $\to k = N^2$;*
- Sector coupling *(α_s): pure* SU(2) *directions ($N = 3$) carry no color charge $\to$ $k = |\mathrm{PG}| - N = N^2{+}1 = 10$;*
- Ratios *($\sin^2\theta_W$): no correction (numerator and denominator correct at the same rate);*
- Resolvent *($1/\alpha_{\mathrm{em}}$): all-orders calculation, separate framework.*

The decomposition $|\mathrm{PG}| = N + (N^2{+}1) = 3 + 10 = 13$ separates pure-weak from non-weak directions. The rule determines k from the physics *of each quantity (which PG directions couple to it), not from numerical scanning.*

Strong CP solution. The potential $V(\phi) = V(-\phi)$ is exactly CP-symmetric: $\phi \to -\phi$ corresponds to $P \to 1-P$ (particle–antiparticle interchange in occupation space). Since the fundamental Lagrangian inherits this symmetry, $\theta_{\text{QCD}} = 0$ identically—the strong CP problem does not arise. No axion is needed. Experiment: $|\theta_{\text{QCD}}| < 10^{-10}$. Prediction: $\theta_{\text{QCD}} = 0$ (exact). ✓

CP violation in the CKM sector. $V(\phi) = V(-\phi)$ implies exact CP at leading order ($V_{ub} = 0$, Jarlskog $J = 0$). The observed CKM CP violation ($J = 3.18 \times 10^{-5}$) arises at higher order: the Wolfenstein structure $J \propto \varepsilon^6$ gives $J \sim 10^{-4}$, the correct order of magnitude. Strong CP (θ_{QCD}) is protected by the *exact* $\phi \leftrightarrow -\phi$ symmetry of $V = -H$; weak CP (δ_{CKM}) is not, because the domain-wall embedding (Section 7.3) breaks this symmetry through the γ^5 coupling.

6.2 Mass formula

The virial-corrected mass formula involves three eigenstate quantities—energy, localization, and the virial ratio:

$$m_n \propto |E_n| \times \text{IPR}_n^{(N^2-1)/N} \times \left(\frac{\langle H \rangle_n}{T_n}\right)^{(N-n_{\text{WKB}})/N}, \tag{6}$$

where $\text{IPR}_n = \int |\psi_n|^4 \, d\phi$ (inverse participation ratio), $T_n = \frac{1}{2}\int \psi_n'^2 \, d\phi$ (gradient energy—a wavefunction property, not the BenDaniel–Duke [15] kinetic energy), $\langle H \rangle_n = \int \psi_n^2 \, H \, d\phi$ (entropy expectation). The IPR exponent $(N^2-1)/N = 8/3 = 2C_F$, where $C_F = (N^2-1)/(2N)$ is the fundamental Casimir. The virial exponent $(N-n_{\text{WKB}})/N = \varepsilon/N = 0.073$ is determined by the WKB deficit of $V = -H$. All quantities are fixed by the spectrum; no free parameters remain.

Both exponents are determined by two independent constraints:

(i) *Bilinear structure*: mass is a bilinear $\bar{\psi}\psi$ (two fermion legs, each carrying C_F), giving IPR exponent $= 2C_F = (N^2-1)/N$. In $d = 0$ the loop integral is absent, so the anomalous dimension is pure group theory. The IPR serves as the renormalization scale: localized states (high IPR) receive larger gauge corrections (established in Anderson localization [16]).

(ii) *Koide equipartition*: $Q = 3/2$ (Section 5). Given $b = 2C_F$, the requirement $Q = 3/2$ uniquely fixes $c = 0.0730$, matching $\varepsilon/N = 0.0731$ to 0.06%. No other value of b produces $c \approx \varepsilon/N$ under the $Q = 3/2$ constraint:

b	$c\,(Q{=}3/2)$	$\lvert c - \varepsilon/N \rvert/(\varepsilon/N)$
2.0	0.437	500%
2.5	0.163	120%
$\mathbf{8/3 = 2C_F}$	**0.0730**	**0.06%**
2.78	0.011	85%
3.0	−0.107	246%

The exponents are not fitted to mass data; $R = 1.5293$ is a *prediction.*

	Eq. (6)	Experiment	Accuracy
m_τ/m_e	3481	3477	0.1%
m_μ/m_e	207	207	0.0%
R	1.5293	1.5294	0.007%
Q (Koide)	1.500	1.500	$< 0.01\%$

$R \in [1.44, 1.57]$ for all power-law forms: a parameter-free bound.

Comparison with the IPR-only formula. The simpler formula $m_n \propto |E_n| \cdot \mathrm{IPR}_n^{n_{\mathrm{WKB}}}$, with $n_{\mathrm{WKB}} = 2.781$, gives $R = 1.519$ (0.7%) and $Q = 1.503$ (0.2%)—good but inferior. The virial-corrected formula (6) improves R by a factor of 100 and reproduces $Q = 3/2$ exactly, because the total exponent splits as $\dim \mathrm{SU}(N)/N + \varepsilon/N = (N^2-1)/N + (N-n_{\mathrm{WKB}})/N$, with the Casimir capturing localization and the WKB deficit capturing virial anharmonicity.

Convergence. All spectral quantities—eigenvalues, IPR, virial ratios, and the resolvent trace (21)—are computed with $|\phi|_{\max} = 50$, $\Delta\phi = 0.0016$ ($N_{\mathrm{grid}} = 64001$), and verified stable under doubling of grid resolution ($|\psi_{\mathrm{edge}}| < 10^{-7}$). At $|\phi|_{\max} = 25$: E_2 is 0.8% too low, R degrades to 1.536 (0.4%), and $1/\alpha_{\mathrm{em}}$ shifts to 137.78 (0.5%). At $|\phi|_{\max} = 100$: $R = 1.529$ and $1/\alpha_{\mathrm{em}} = 137.035$, confirming convergence at $|\phi|_{\max} = 50$.

6.3 Up-type quarks: $c = N_c = 3$

The color number $N_c = 3$ is derived, not assumed: $d = 3$ (Theorem 4) $\to \nabla\phi \in \mathbb{C}^3 \to \mathrm{CP}^2 \to \mathrm{SU}(3)_{\mathrm{color}} \to N_c = 3$. Color confinement forces all N_c color modes to share a single source field ϕ (because confined quarks cannot be isolated; their occupation parameters are locked together). In the 0D QFT, N_c identical modes give

$$Z_{N_c}(\phi) = (1 + e^{\phi})^{N_c}, \quad W_{N_c} = N_c \ln(1+e^{\phi}), \quad \Gamma_{N_c} = N_c\,\Gamma_1 = -N_c\,H(\sigma). \tag{7}$$

Hence $V = -N_c H = -cH$ with $c = N_c$: the potential depth is determined by the color number, not fitted. At $c = 3$: $N = 5$ bound states, $n_{\mathrm{WKB}} = 4.82$.

With the same exponents $b = 2C_F$, $c = \varepsilon/N$ as the lepton sector (§6.2) and modes $(1, 2, 3)$: $R_{\mathrm{up}} = 1.777$ (experiment: 1.772, deviation 0.3%). Both $c = 3$ and $b = 2C_F$ are derived quantities. This is a zero-parameter prediction.

6.4 Down-type quarks: effective mass from weak isospin

At $c = 3$ with constant mass, modes $(0, 1, 2)$ give $R_{\mathrm{down}} = 2.096$ (experiment: 2.269, 7.6% off). The discrepancy is resolved by a variable effective mass.

In the symmetric sector $\phi_r = \phi_g = \phi_b = \phi$, fluctuations off the diagonal induce an effective mass correction proportional to the Fisher metric $g_F = \sigma(1-\sigma)$, which is maximal at $P = 1/2$ (maximum uncertainty) and vanishes at $P = 0, 1$:

$$m_{\mathrm{eff}}(\phi) = 1 + g\,\sigma(1-\sigma). \tag{8}$$

The coupling g depends on weak isospin I_3:

$$g = 0 \;\; (I_3 = +\tfrac{1}{2}), \qquad g = -g_0 \;\; (I_3 = -\tfrac{1}{2}), \tag{9}$$

with

$$g_0 = 2C_F + \frac{1}{N_c} = \frac{N_c^2-1}{N_c} + \frac{1}{N_c} = \frac{N_c^2}{N_c} = N_c = 3. \tag{10}$$

The vanishing of g for $I_3 = +1/2$ ensures that the up-quark R-value remains parameter-free. The identity $2C_F + 1/N_c = N_c$ is exact: $2C_F = (N_c^2-1)/N_c$ is the SU(2) doublet's total color Casimir, and $1/N_c$ is the baryon U(1) charge. A2 assigns the entire doublet color charge to the contrast component ($I_3 = -1/2$).

With $g = -N_c = -3$ and modes $(0, 1, 2)$: $R_{\text{down}} = 2.273$ (experiment: 2.269, deviation 0.2%). Using the BenDaniel–Duke kinetic energy $T_{\text{BD}} = \frac{1}{2}\int Z_2\, \psi'^2\, d\phi$ instead of $T = \frac{1}{2}\int \psi'^2\, d\phi$ gives $R_{\text{down}} = 2.30$ (1.5%); the wavefunction-only definition is used throughout to avoid double-counting ($|E_n|$ already encodes m_{eff}).

Sector	c	g	R (predicted)	R (experiment)
Lepton	1	0	1.529	1.529 (0.007%)
Up	$N_c = 3$	0	1.777	1.772 (0.3%)
Down	$N_c = 3$	$-N_c$	2.273	2.269 (0.2%)

All three R-values are reproduced within $\sim$1%. All ingredients are derived: $c = N_c = d = 3$ (Theorem 4 $\to$ SU(3) $\to$ $N_c = 3$; confinement $\to$ $c = N_c$); $g_0 = 2C_F + 1/N_c = N_c$ (doublet Casimir + baryon U(1)); mode assignments up $= (1, 2, 3)$, down $= (0, 1, 2)$ (at $c = N_c$, mode 0 is deeply confined below the generation window; the contrast component with $g = -N_c$ has $N = 4$ states, and $(0, 1, 2)$ are the three lightest).

Derivation of g_0. The SU(2) doublet (u, d) carries total color Casimir $2C_F = (N_c^2-1)/N_c = 8/3$, plus baryon U(1) charge $1/N_c = 1/3$. The A2 contrast principle assigns the *entire* doublet charge to the $I_3 = -1/2$ component:

$$g_0 = 2C_F + \frac{1}{N_c} = \frac{N_c^2-1}{N_c} + \frac{1}{N_c} = \frac{N_c^2}{N_c} = N_c = 3. \tag{11}$$

This identity is exact for all N_c. The result $g_0 = c = N_c$ means a single quantity—the number of confined color modes—determines both the potential depth ($V = -N_c H$) and the mass correction ($m_{\text{eff}} = 1 - N_c\, \sigma(1-\sigma)$).

6.5 Neutrino mass prediction

The ratio R is universal for all fermions coupled to $V = -H$ with $c = 1$: it depends only on the eigenstate properties of the potential, not on the absolute mass scale. For Dirac neutrinos, $R_\nu = R_{\text{lepton}} = 1.529$. The seesaw mechanism $m_\nu = m_D^2/M_R$ preserves R exactly if $M_R \propto \mathbb{1}$ (universal right-handed mass), since $R = \ln(m_0^2/m_2^2)/\ln(m_1^2/m_2^2) = R_{\text{Dirac}}$. Combined with oscillation data ($\Delta m_{21}^2 = 7.53\times10^{-5}$ eV^2, $\Delta m_{31}^2 = 2.45\times10^{-3}$ eV^2, normal ordering), $R_\nu = 1.529$ determines m_1 uniquely:

	Prediction	Status
m_1	0.32 ± 0.03 meV	unmeasured
m_2	8.68 meV	consistent
m_3	49.5 meV	consistent
$\sum m_i$	58.5 ± 0.3 meV	below Planck (< 120)
Ordering	normal	JUNO (2026–27)

The sum $\sum m_i = 58.5$ meV is dominated by $m_3 \approx \sqrt{\Delta m_{31}^2} = 49.5$ meV; the theory's contribution is $m_1 \approx 0.3$ meV, placing $\sum m_i$ near its minimum allowed value. The Koide parameter $Q_\nu = 1.90 \neq 3/2$: neutrinos do not satisfy the equipartition condition, consistent with the theory's prediction that $Q = N/2$ holds only for charged leptons (all $N = 3$ modes used).

Falsifiability. (i) Inverted ordering $\to$ theory falsified. (ii) $\sum m_i > 70$ meV (requiring $R > 1.8$) $\to$ tension. (iii) CMB-S4 sensitivity ($\sigma \sim 15$ meV) can test $\sum m_i = 58.5$ meV directly.
Caveat. A generic seesaw matrix $M_R \not\propto \mathbb{1}$ modifies R_ν; the prediction assumes either Dirac masses or a flavor-universal M_R.

6.6 Higgs mass ratio

The potential $V = -H$ has curvature at the center: $V''(0) = g_F(0) = \sigma(0)(1-\sigma(0)) = 1/4$, the Fisher metric at maximum uncertainty ($P = 1/2$). In QFT, $m^2 = V''(\phi_{\min})$ gives the mass of small oscillations:

$$\frac{m_H}{v} = \sqrt{V''(0)} = \frac{1}{2} \qquad \text{(experiment: } 0.509,\ 1.7\%\text{).} \tag{12}$$

This corresponds to $\lambda = 1/8 = 0.125$ (experiment: 0.129, 3.4%), consistent with radiative corrections at tree level.

Status of m_H/v. Eq. (12) identifies $V = -H$ with the Higgs potential and $\phi = 0$ with the electroweak vacuum. The physical Higgs potential $V_H = \mu^2|\Phi|^2 + \lambda|\Phi|^4$ has $m_H/v = \sqrt{2\lambda}$; Eq. (12) gives $\lambda = 1/8$. Whether $V = -H$ is literally the Higgs potential or merely shares its curvature is an open question.

6.7 Neutrino mixing angles

CKM mixing is perturbative ($\sin\theta_C \approx \varepsilon \ll 1$): the internal coordinate ϕ controls mixing. PMNS mixing is non-perturbative ($\sin^2\theta_{12} \approx 1/3$): the gauge geometry controls mixing.

A line in $\mathrm{PG}(2,3)$ contains $N+1 = 4$ points. The solar mixing angle measures the fraction of PG covered by one complete electroweak line ($\mathrm{SU}(2) \times \mathrm{U}(1)$, including hypercharge):

$$\sin^2\theta_{12} = \frac{N+1}{|\mathrm{PG}|} = \frac{4}{13} = 0.3077 \qquad \text{(experiment: } 0.307,\ 0.2\%\text{).} \tag{13}$$

Compare $\sin^2\theta_W = N/|\mathrm{PG}| = 3/13$ (pure SU(2), excluding hypercharge); the difference $1/|\mathrm{PG}|$ is the $\mathrm{U}(1)_Y$ contribution.

The atmospheric mixing is maximal (μ–τ symmetry) plus a correction at the PG scale:

$$\sin^2\theta_{23} = \frac{1}{2} + \frac{1}{2|\mathrm{PG}|} = \frac{7}{13} = 0.538 \qquad \text{(experiment: } 0.546,\ 1.4\%\text{).} \tag{14}$$

The correction $1/(2|\mathrm{PG}|)$ is the minimum μ–τ symmetry breaking from the gauge geometry; the sign ($> 1/2$) follows from τ being heavier (more localized, larger PG overlap). Consistency check: $\sin^2\theta_{23} = \sin^2\theta_W + \sin^2\theta_{12}$ ($7/13 = 3/13 + 4/13$): atmospheric mixing exhausts both the pure-weak (N points) and the full electroweak-line ($N+1$ points) fractions of $\mathrm{PG}(2,3)$. The reactor angle is suppressed by the full PG: $\sin^2\theta_{13} = 1/(N \cdot |\mathrm{PG}|) = 1/39 = 0.026$ (experiment: 0.022, 17%; Tier C).

Status of PMNS. CKM involves the internal coordinate ϕ ($\varepsilon \ll 1$, perturbative); PMNS involves the gauge geometry ($1/N \sim 1$, non-perturbative). $\theta_{12} = 4/13$ is a PG line fraction (Tier B); $\theta_{23} = 7/13$ follows from μ–τ breaking at the PG scale (Tier B).

7 Connection to Four-Dimensional Physics

7.1 Two geometries from $H(P)$

Internal geometry. $H(P) \to V = -H$ (canonical free energy); $\phi = \mathrm{logit}(P)$ (natural parameter) $\to$ Schrödinger equation.

Spacetime (Einstein). The action $S[\phi, g_{\mu\nu}]$ depends on the metric $g_{\mu\nu}$. Background independence requires $g_{\mu\nu}$ to be dynamical: fixing $g_{\mu\nu} = \eta_{\mu\nu}$ would be a structural choice not determined by the theory. Lovelock's theorem [8] then guarantees that Einstein's equation $G_{\mu\nu} + \Lambda g_{\mu\nu} = \kappa T_{\mu\nu}$ is the *unique* gravitational dynamics in $d{+}1 = 4$ dimensions (in $d{+}1 \geq 6$, Gauss–Bonnet terms appear; the derivation of $d = 3$ (Theorem 4) eliminates them). Independently, Jacobson [7]: $\delta Q = T\,dS$ on causal horizons $\to$ Einstein's equation. If $S = H(P)$ fundamentally, both routes—action-based (Lovelock) and thermodynamic (Jacobson)—yield Einstein gravity from binary entropy.

7.2 Domain-wall embedding in 4+1D

The internal–external decomposition $\Psi(x, \phi) = \sum_n \psi_n(\phi)\,\chi_n(x)$ is the standard domain-wall fermion mechanism [11] applied to $V = -H$. In 4+1 dimensions with coordinates (x^μ, ϕ):

$$S_{5D} = \int d^4x\, d\phi\; \bar{\Psi}\big[\gamma^\mu \partial_\mu + \gamma^5 \partial_\phi + V(\phi)\big]\Psi, \tag{15}$$

the potential $V = -H$ traps $N = 3$ bound states on the wall, each giving a 3+1D fermion $\chi_n(x)$. Chirality arises from γ^5: left- and right-handed modes localize at $\phi < 0$ ($P < 1/2$) and $\phi > 0$ ($P > 1/2$). The 4D mass m_n is determined by the overlap of left and right profiles at $\phi = 0$, proportional to $|\psi_n(0)|^2$. The mass hierarchy is automatic: deeply bound modes (large $|\psi_n(0)|^2$) are heavy; delocalized modes (small $|\psi_n(0)|^2$) are light. The 5D gauge field decomposes via $\mathrm{PG}(2,3)$, and the 4D coupling $1/g_{4D}^2 = \int d\phi\, |A_\phi|^2$ recovers the resolvent trace of Section 8. The spatial dimension $d = 3$ is not assumed but derived (Theorem 4), fixing the wall's codimension to 1.

8 Gauge Structure from the Projective Plane

8.1 Why $\mathrm{PG}(2,3)$: the $N^N = N^3$ theorem

Projective geometry enters as a consequence of gauge invariance. Gauge transformations act linearly on N internal states (standard); the overall phase is unobservable (gauge invariance); in $d = 3$ spatial dimensions, the gradient $\nabla\phi$ has d components; the independent gauge directions are therefore the elements of $\mathbb{F}_N^d \setminus \{0\}$ modulo scalar multiplication $= \mathrm{PG}(d{-}1, \mathbb{F}_N)$. With $d = N = 3$: $\mathrm{PG}(2, \mathbb{F}_3)$, $|\mathrm{PG}| = (N^d{-}1)/(N{-}1) = 13$ points, 52 flags. This is not a postulate but a consequence of linearity + gauge invariance + $d = N = 3$.

Why $\mathbb{F}_3$, not $\mathbb{C}^3$. The Sturm–Liouville theorem guarantees that the bound-state energies E_0, E_1, E_2 are non-degenerate. A non-degenerate Hamiltonian has a preferred eigenbasis; its commutant consists of diagonal operators only (Schur). Gauge symmetry, being exact, must commute with H and therefore preserve the eigenbasis—acting as *permutations* of the mass eigenstates, not continuous rotations. Linear permutations on N labels require a field with $|K| = N$; since $N = 3$ is prime, $K = \mathbb{F}_3$ is unique. CKM mixing ($\sin\theta_C = \varepsilon \ll 1$) is a perturbative off-diagonal correction that does *not* commute with H and is therefore a physical effect, not a gauge symmetry. The contrast with QCD is instructive: quark colors are *degenerate* (equal masses), so no preferred basis exists and the gauge group is continuous (SU(3) over $\mathbb{C}$); generations are *non-degenerate*, so the eigenbasis is preferred and the gauge structure is discrete ($\mathrm{GL}(d, \mathbb{F}_3)$ over $\mathbb{F}_3$).

When $d = N$, the projective structure is *self-referential.*

Theorem 6 (Self-referential uniqueness)**.** *Let $V = -H(\sigma(\phi))$ with unit mass in ϕ, and let $N(d)$ denote the number of bound states of the s-wave equation in d spatial dimensions, $-\frac{1}{2}u'' + [V(r) + \frac{(d-1)(d-3)}{8r^2}]u = Eu$ (Section 4). The following conditions are individually necessary and jointly satisfied* only *by $N = 3$:*

(i) Fixed point*: $N(d) = d$ (generation count equals spatial dimension);*

(ii) Algebraic*: $N^N = N^3$ (self-referential exponent);*

(iii) Geometric*: $|\mathrm{PG}(N{-}1, N)| = N^2 + N + 1$ (projective plane has polynomial size).*

Proof. (i) $N(1) = 3$, $N(2) = 3$, $N(3) = 3$, $N(4) = 2$, $N(d \geq 5) = 0$ (numerical; Theorems 1–4). The unique solution is $d = 3$. (Note: $N(d)$ counts bound states of $V = -H$ with a centrifugal correction; this is distinct from the color-scaled equation $V = -N_c H$ in Section 6, which has 5 bound states.) The physical content of $N(d) = d$: each generation "fills" one spatial dimension; excess generations ($N > d$) destabilize, and excess dimensions ($d > N$) are unoccupied. (ii) $N^N = N^3$ iff $N^{N-3} = 1$; for $N \geq 2$ this requires $N = 3$. (iii) $|\mathrm{PG}(N{-}1, N)| = (N^N{-}1)/(N{-}1) = N^2{+}N{+}1$ iff $N^N = N^3$, i.e. $N = 3$ by (ii). □

The three conditions have independent origins: (i) from the spectrum of $V = -H$, (ii) from pure algebra, (iii) from finite projective geometry. Their simultaneous satisfaction at $N = 3$ means that the number of fermion generations, the spatial dimension, and the gauge structure of the Standard Model are not independent parameters but consequences of a single self-referential constraint.

Remark 5 (Why $d = N$)**.** *In most physical theories d and N are independent. Here the fixed-point equation $N(d) = d$* forces *$d = N = 3$. The projective plane* $\mathrm{PG}(d{-}1, N) = \mathrm{PG}(N{-}1, N)$ *is then self-referential: the field ($\mathbb{F}_N$) and the ambient dimension (d) share the same number. This self-referential structure exists only for $N = 3$.*

The decomposition $13 = 9 + 3 + 1 = N^2 + N + 1$ mirrors the gauge group $\mathrm{U}(N) \times \mathrm{SU}(2) \times \mathrm{U}(1)$: $N^2 = \dim \mathrm{U}(N)$, $N = \dim_{\text{fund}}$, $1 = \dim \mathrm{U}(1)$.

8.2 Electroweak breaking as a geometric choice

In $\mathrm{PG}(2, N)$, a *line* contains $N + 1 = 4$ points. Choosing a point P_0 (hypercharge) and a line L through it partitions the 13 points:

$$\underbrace{N^2}_{\text{9 color}} + \underbrace{N}_{\text{3 weak}} + \underbrace{1}_{\text{1 hyp}} = N^2 + N + 1. \tag{16}$$

The $N = 3$ points on $L \setminus \{P_0\}$ are the weak directions; the $N^2 = 9$ points in $\mathrm{PG} \setminus L$ are the color sector. Before symmetry breaking, all 13 directions are equivalent (GUT); the choice of (P_0, L) selects the Standard Model subgroup.

8.3 Gauge couplings

Gauge couplings from the invariant measure. The automorphism group $\mathrm{PGL}(3, \mathbb{F}_3)$ acts *transitively* on $\mathrm{PG}(2, 3)$: any point can be mapped to any other. The unique probability measure invariant under a transitive group action is uniform (Haar measure). Before symmetry breaking, the gauge-invariant weight of each direction is therefore $1/|\mathrm{PG}| = 1/13$—not a postulate but a theorem. The Weinberg angle [18] is the fraction of weak directions:

$$\sin^2\theta_W = \frac{|\text{weak}|}{|\mathrm{PG}|} = \frac{N}{N^2+N+1} = \frac{3}{13} = 0.2308. \tag{17}$$

W/Z **mass ratio.** The Weinberg angle determines the W/Z mass ratio at tree level:

$$\frac{m_W}{m_Z} = \cos\theta_W = \sqrt{1 - \frac{3}{13}} = \sqrt{\frac{10}{13}} = 0.8771 \qquad \text{(experiment: } 0.8815,\ 0.5\%\text{)}. \tag{18}$$

The 0.5% deviation is consistent with radiative corrections ($\Delta\rho \sim \alpha/\pi$).

Strong coupling from the torus. The automorphism group $\mathrm{PGL}(d, \mathbb{F}_N)$ has maximal torus of dimension $\dim T = d-1 = N-1 = 2$. The tree-level strong coupling per torus direction is $\alpha_s^{(0)} = \sin^2\theta_W/\dim T = 3/26 = 0.1154$. The $|\mathrm{PG}| = N^2+N+1 = 13$ gauge directions decompose as $N = 3$ weak ($\mathrm{SU}(2)$) and $N^2+1 = 10$ non-weak. The marginality correction (Remark 4) acts through the $k = N^2+1 = 10$ non-weak PG directions:

$$\alpha_s = \frac{3}{26}\left(1 + \frac{\varepsilon}{N^2+1}\right) = 0.1179 \qquad \text{(experiment: } 0.1179,\ 0.01\%\text{)}. \tag{19}$$

Fine-structure constant from linear response. In 4D, $1/\alpha$ is determined by vacuum polarization—a closed fermion loop with two propagators ($\mathrm{Tr}[G^2]$). In 0D there is no spacetime for pair creation; the gauge coupling is instead the *linear susceptibility* of the internal space to gauge field fluctuations:

$$\chi = -\frac{\partial^2 F}{\partial A^2}\bigg|_{A=0} = \mathrm{Tr}\left[\frac{1}{K_{\text{int}}}\right], \tag{20}$$

where $K_{\text{int}} = K_{\text{gen}} \otimes 1 + 1 \otimes \Delta_{\text{flag}}$ is the total internal kinetic operator (generation binding energies $|E_n|$ plus flag Laplacian λ_k). This is a theorem of linear response theory: the susceptibility is the resolvent trace ($s = 1$), not $\mathrm{Tr}[1/K^2]$ ($s = 2$, which would give $1/\alpha \approx 9258$). The resolvent gives $Z = 134.78$. Adding the spectral gap $\lambda_1 = 4 - \sqrt{3}$ (the minimum gauge transition energy—the only physically motivated IR scale) and imposing Dyson self-consistency (one photon channel feeds back into the coupling):

$$\frac{1}{\alpha} + \alpha = Z + \lambda_1. \tag{21}$$

Solution: $1/\alpha = [(Z+\lambda_1) + \sqrt{(Z+\lambda_1)^2 - 4}]/2 = 137.037$ (experiment: 137.036, 0.0009%).

The resolvent trace decomposes as $Z = \zeta_V(1) + 12\, S_1 + 12\, S_2 + 27\, S_3$, where $S_i = \sum_n (E_n + \lambda_i)^{-1}$ and the multiplicity $27 = N^N$ (Theorem 6). The N^N sector contributes $9.9 \approx N^2$, providing a spectral origin for the QCD vacuum-polarization correction. Only bound states enter; continuum states ($E \to 0^+$) are excluded because only physical generations contribute to vacuum polarization.

Experimental values: $\sin^2\theta_W = 0.2312$ (0.2%), $\alpha_s(M_Z) = 0.1179$ (0.01%), $1/\alpha_{\text{em}} = 137.036$ (0.0009%). All three gauge couplings are derived from $V = -H$ and $\mathrm{PG}(2, 3)$.

8.4 Flag Laplacian spectrum

A flag in $\mathrm{PG}(2,N)$ is a point-on-line incidence pair; flags are the natural basis for the gauge Laplacian because gauge fields propagate along incidence relations, not just points.

The flag variety of $\mathrm{PG}(2,N)$ has $(N^2{+}N{+}1)(N{+}1) = 52$ elements (point–line incidence pairs). The flag graph (adjacency when two flags share a point or line) has degree $2N = 6$ and Laplacian spectrum:

Eigenvalue	Multiplicity	
0	1	trivial
$(N{+}1) - \sqrt{N}$	$N^2{+}N$	$= 12$
$(N{+}1) + \sqrt{N}$	$N^2{+}N$	$= 12$
$2(N{+}1)$	N^N	$= 27$

The lowest non-trivial eigenvalue $\lambda_1 = (N{+}1) - \sqrt{N} = 4 - \sqrt{3} = 2.268$ matches $R_{\mathrm{down}} = 2.272$ to 0.17%. The highest eigenvalue has multiplicity N^N, which equals $N^3 = 27$ only for $N = 3$ (Theorem 6). Whether the near-equality $\lambda_1 \approx R_{\mathrm{down}}$ reflects a duality between the continuous spectrum of $V = -H$ and the discrete spectrum of the flag Laplacian is an open question.

8.5 Spectral participation ratio

The bound-state density $\rho(\phi) = \sum_{n=0}^{N-1} |\psi_n(\phi)|^2$ has participation ratio

$$D_{\mathrm{PR}} = \frac{(\int \rho\, d\phi)^2}{\int \rho^2\, d\phi} = \frac{N^2}{\sum_{nm} C_{nm}}, \qquad C_{nm} = \int |\psi_n|^2 |\psi_m|^2\, d\phi. \tag{22}$$

For $V = -H$ ($c = 1$): $D_{\mathrm{PR}} = 13.06 = |\mathrm{PG}(2,3)|$ to 0.4%. Among the family $V = -cH$, $D_{\mathrm{PR}}(c)$ is monotonically decreasing and passes through 13 near $c = 1$: $D_{\mathrm{PR}}(0.95) = 13.8$, $D_{\mathrm{PR}}(1.00) = 13.06$, $D_{\mathrm{PR}}(1.05) = 12.4$. The derivation chain thus has two independent paths from $V = -H$ to $\mathrm{PG}(2,3)$: the eigenvalues determine $N = 3$ (Theorem 1); the eigenfunctions determine $D_{\mathrm{PR}} \approx |\mathrm{PG}| = 13$.

Remark 6 (Status of gauge couplings)**.** *All three gauge couplings are derived from $V = -H$ and $\mathrm{PG}(2,3)$. Equations (17)–(19) follow from the PGL-invariant measure and the torus of PGL. Equation (21) is a self-consistent Dyson equation: $1/\alpha_{\mathrm{tree}} = Z + \lambda_1$ is the tree-level coupling, and the self-screening correction $-\alpha$ accounts for the photon's own contribution to vacuum polarization. The coefficient of α is 1 because there is exactly one U(1) photon channel and the one-loop self-energy is first order in the coupling; $k = 0$ (no screening) gives $1/\alpha = 137.044$ (0.006% off), $k = 2$ gives 68.5 (factor-of-2 wrong). The formula selects $\alpha_{\mathrm{em}}(0)$ because the resolvent trace sums all channels without momentum cutoff, corresponding to the fully screened long-distance limit. Adding q^2 to the resolvent denominators yields $\alpha(q^2)$ that increases with energy, consistent with QED. The residual 0.001% deviation is consistent with $O(\alpha^2)$ corrections beyond tree level.*

Remark 7 (Look-elsewhere effect)**.** *A look-elsewhere analysis was performed: 2226 algebraic expressions were generated from $\{N, N^2{+}N{+}1, N^2, N{\pm}1, \pi, \pi^2\}$ using $\{+, -, \times, /\}$. The fraction matching each target within the observed accuracy: $\sin^2\theta_W$ within 0.5%: $3/2226 = 0.13\%$; α_s within 2.5%: $6/2226 = 0.27\%$; $1/\alpha_{\mathrm{em}}$ within 0.5%: $4/2226 = 0.18\%$. The joint probability is $\lesssim 10^{-7}$ if independent. Among integers $N = 2, \ldots, 10$, only $N = 3$*

reproduces all three couplings simultaneously (combined error 2.2%; next best $N = 4$ at 80%). Since $N = 3$ is fixed by the spectrum of $V = -H$ (Theorem 1), it is not a free parameter. The low joint probability is necessary but not sufficient for establishing a physical connection: without a derivation, the relations remain numerology, however statistically suggestive.

9 Derivation Status

The derivation chain from A1–A3 to the Schrödinger equation involves four steps, each with independent justification: (i) $V = \Gamma = -H$ is the exact 0D effective potential (Legendre transform); (ii) $Z_2 = 1$ follows from the flat path-integral measure in ϕ (exponential-family structure, not a choice); (iii) the transfer matrix of the canonical action *is* the Schrödinger equation (standard lattice QFT); (iv) $d = 3$ is the unique dimension preserving $N = 3$ with spin-statistics and propagating gravity. The identification $\mathbb{F}_3$ (not $\mathbb{C}^3$) follows from non-degeneracy (Sturm–Liouville $\to$ Schur $\to$ permutations $\to$ $N = 3$ prime $\to$ $\mathbb{F}_3$ unique); $D_{\text{PR}} = 13.06 \approx |\text{PG}(2, 3)|$ provides independent evidence. Self-consistency: $N = 3$ describes energy *levels* of one mode ($Z = 1 + e^{\phi}$), not three independent fields.

The quark sector introduces one physical input (confinement: $\phi_r = \phi_g = \phi_b$) yielding $c = N_c = 3$ and $R_{\text{up}} = 1.777$ (0.3%). A2's reference–contrast structure assigns $g = 0$ ($I_3 = +1/2$) and $g = -N_c$ ($I_3 = -1/2$), giving $R_{\text{down}} = 2.273$ (0.2%). The mass formula exponents are determined by two constraints: $2C_F$ (bilinear $\bar{\psi}\psi$) and ε/N (Koide $Q = N/2$); no other IPR exponent satisfies both. Internal–external decoupling $\Psi(x, \phi) = \sum_n \psi_n(\phi)\chi_n(x)$ is the domain-wall mechanism—a change of variables, and the theory's largest open assumption.

The 0D theory computes dimensionless internal-structure parameters in three categories: (A) scale-independent quantities (mass ratios, mixing angles, Q, θ_{QCD}); (B) the IR coupling $1/\alpha = 137.037$ (resolvent at $q^2 = 0$, Thomson limit); (C) tree-level gauge ratios ($\sin^2\theta_W = 3/13$, α_s), where 3/13 is a counting fraction on $\text{PG}(2, 3)$ with no intrinsic scale (proximity to M_Z rather than $q^2 = 0$ suggests PG matches the SM at the electroweak scale). Spacetime enters only through the absolute mass scale (undetermined by A2) and RG running. Genuinely open: the Higgs mass, CKM CP phase, and absolute mass scale remain outside the framework.

Tier A: Mathematical theorems (from $V = -H$ alone).

Quantity	Prediction	Experiment	Proof
N (generations)	3	3	Variational + Sturm
d (spatial dim.)	3	3	Centrifugal + fixed point
Q (Koide)	3/2	3/2	Equipartition (A2)
θ_{QCD}	0	$< 10^{-10}$	$V(\phi) = V(-\phi)$
$V_{ub} \approx 0$	10^{-13}	0.004	Parity selection
4th generation	forbidden	excluded	$E_3 > 0$

Tier B: QFT derivations ($V = -H$ + $\text{PG}(2, 3)$ + standard QFT).

Quantity	Prediction	Experiment	Accuracy
$1/\alpha_{\rm em}$	137.037	137.036	0.001%
$R_{\rm lepton}$	1.5293	1.5294	0.007%
α_s	0.1179	0.1179	0.01%
$\sin\theta_C$	0.2245	0.2244	0.07%
m_τ/m_e	3481	3477	0.1%
m_μ/m_e	207	207	0.0%
$R_{\rm up}$	1.777	1.772	0.3%
$R_{\rm down}$	2.273	2.269	0.2%
$\sin^2\theta_W$	3/13	0.2312	0.2%
m_W/m_Z	$\sqrt{10/13}$	0.8815	0.5%
$\sin^2\theta_{12}$	4/13	0.307	0.2%
$\sin^2\theta_{23}$	7/13	0.546	1.4%

Tier C: Structural correspondences (correct order, less precise).

Quantity	Prediction	Experiment	Accuracy
m_H/v	1/2	0.509	1.7%
$\sin^2\theta_{13}$	1/39	0.022	17%
$\sum m_\nu$	58.5 meV	< 120	testable

All predictions follow from one potential ($V = -H$) and one geometric structure ($\mathrm{PG}(2,3)$). The Standard Model's 19+ parameters are compressed to one (the absolute mass scale, undetermined by design). Falsifiable predictions include: normal neutrino ordering (JUNO), $\sum m_\nu = 58.5$ meV (CMB-S4), and fourth generation absolutely forbidden.

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